STATEMENT OF NEED
OSEL Procurement Request Form
SON Optical Coherence Tomography Retinal Imaging System
January 24, 2022

1. BACKGROUND

The Food and Drug Administration (FDA) is responsible for protecting the public health by assuring the safety, efficacy, and security of human and veterinary drugs, biological products, medical devices, our nation's food supply, cosmetics, and products that emit radiation. The FDA is also responsible for advancing the public health by helping to speed innovations that make medicines and foods more effective, safer, and more affordable, and helping the public get the accurate, science-based information they need to use medicines and foods to improve their health.

The mission of the Center for Devices and Radiological Health (CDRH) is to protect and promote the public health. CDRH assures that patients and providers have timely and continued access to safe, effective, and high-quality medical devices and safe radiation-emitting products. We provide consumers, patients, their caregivers, and providers with understandable and accessible science-based information about the products we oversee. We facilitate medical device innovation by advancing regulatory science, providing industry with predictable, consistent, transparent, and efficient regulatory pathways, and assuring consumer confidence in devices marketed in the U.S.

The Office of Science and Engineering Laboratories (OSEL) supports CDRH's mission of protecting and promoting public health. OSEL undertakes the highest quality science to provide their customers with the best methods, tools and expertise to:

- Ensure readiness for emerging and innovative medical technologies
- Develop appropriate evaluation strategies and testing standards
- Create accessible and understandable public health information and,
- Deliver timely and accurate decisions for products across their life cycle.

Under OSEL, the Division of Biomedical Physics (DBP) participates in the Center's mission of protecting and promoting public health by identifying and investigating the biophysical interactions between medical devices and the human body. The division accomplishes this through activities supporting the OSEL mission.

DBP investigates gender-based differences in cardiac resynchronization therapy for pacemakers, development of phantoms (physical models) to assess the measurement of eye disease by optical imaging systems, evaluation and improvement of electrode reliability in neural prosthetics used to control artificial limbs, and computational modeling of active and passive implants to determine if unsafe levels of heating arise during a patient's exposure to magnetic resonance imaging (MRI) systems. These serve the Center's mission of advancing regulatory science, facilitating consistent and efficient regulatory pathways, and assuring continued access to safe, effective, and high-quality medical devices.

Specifically, DBP focuses on device issues that involve:

- Biomedical and tissue optics
- Biophysics and electrophysiology

- Electrical engineering
- Functional device performance and human factors, and
- Wireless communication and electromagnetic interference and compatibility

In collaboration with OSEL, the DBP seeks to order a high-resolution ophthalmic adaptive optics (AO) device will be deployed in the Optical Diagnostic Devices Laboratory of FDA/CDRH/OSEL/DBP to support multiple ongoing studies under the ophthalmic devices program charter. The OSEL ophthalmology program has established an AO sub-program focused on characterizing the performance and capabilities of novel optical diagnostic devices for detecting and treating retinal diseases, while developing sensitive biomarkers for early disease detection, including glaucoma—a leading cause of irreversible blindness. This instrument will directly support several critical projects within the program charter, specifically AO biomarker development that will advance regulatory science tools (RSTs) and retinal phantom development to establish testing standards. The device will facilitate collaborative research with the National Eye Institute, University of Maryland Baltimore, and Johns Hopkins University Wilmer Eye Institute. This cutting-edge equipment will enable the development of novel cellular imaging biomarkers and serve as a foundational platform for creating new analytical methods and tools critical to the program's objectives. The acquisition directly fulfills FDA's core mission to foster and accelerate the clinical translation of innovative medical device technologies by providing the necessary infrastructure to evaluate and validate emerging retinal imaging technologies, support the development of regulatory science tools and standards for cellular-level ophthalmic imaging, and facilitate the assessment of novel biomarkers that may be incorporated into future medical device applications.

2. STATEMENT OF NEED

Item: Adaptive Optics (AO) Retinal Imaging System (including device, mouse) and all features (Software). The AO System shall meet the following salient characteristics:

- Hardware
- Technical specifications
  - Methodology: pupillary reflectance imaging and non-contact transscleral flood illumination
  - Light sources Infrared LED and SLD (750 to 890 nm)
  - Lateral resolution in tissue (optical): 2 — 3 um
  - Pixel pitch: 1 um
  - Focusing range in tissue: 300 um
  - Cellular View image field angle: >6.5° x 6.5°
  - Fundus Overview image field angle: 30° x 30°
  - Min. pupil diameter: 4 mm
  - Refractive error adjustment: +15 D to –15 D, continuous
  - Acquisition time: <3s
  - Fixation targets internal and external

- Software

- Simultaneous live widefield overview and cellular-level imaging
- Pupil alignment feedback
- Image comparison for follow-up examination

- Electronic Interfaces and others
  - Input voltage and frequency : 100 — 240 V, 50 — 60 Hz
  - Power consumption (max) : 450 W
  - Dimensions (L x W x H) : <= 65cm x 65cm x 65cm
  - Weight : <= 60 kg

The item(s) are to be delivered to:

Contact program manager (PM):
Attn: Zhuolin Liu

U.S. Food and Drug Administration
CDRH/OSEL/DBP

10903 New Hampshire Avenue
WO62 Room 1124
Silver Spring, MD 20993
Phone: 301-796-7914
Email: Zhuolin.Liu@fda.hhs.gov

3. CONTRACT TYPE

TBD

4. ASSEMBLY AND DELIVERY REQUIRMENTS

General Support Requirements:

This contractor shall provide:

- Initial delivery and setup of system
- User manuals
- Supply all hardware and any associated software with appropriate license keys, expiration dates and proof of entitlement certificates
- Offer training sessions on device operation provided by their company
- Have deliverables required under this contract to be packaged, marked and shipped in accordance with Government specifications
- Have deliverables marked with the contract number and contractor's name
- All required materials must be delivered in immediate new, usable and acceptable condition

- Contact Program Manager (PM) to schedule the delivery. All deliveries must be made during normal FDA White Oak Campus delivery hours: 8:00AM to 3:30PM

5. PERIOD OF PERFORMANCE

TBD

6. PLACE OF PERFORMANCE

U.S. Food and Drug Administration
CDRH/OSEL/DBP
10903 New Hampshire Ave
WO62, Room G101
Silver Spring, MD 20993

7. WARRANTY MAINTENANCE AND TECHNICAL SUPPORT

This contract shall include:

- A full 1 year warranty after delivery

NOTE to Contractor/Vendor: Before the contract is awarded, we are required to get Pre-Approval of all the IT Hardware and/or Software-Firmware-Freeware from the FDA Chief Information Officer (CIO). For IT Hardware, this includes any device that process or stores data or controlled by data (computers/data switches, etc..) but does not include passive hardware (rack, network cables, power supplies/cords). This will require the applicable contractor to provide a complete list of Hardware and/or Software-Firmware-Freeware that the contractor will use in fulfilling this contract.

This list will need to include: IT hardware: Manufacture, nomenclature and model number
Software (all types): Manufacture, nomenclature and version number
Item(s) rejected by the CIO will need to be changed and the replacement item(s) would need to go through the same approval process.

Deliverables will conform to 36 CFR Part 1194.41, "Information, Documentation and Support," and 36 CFR Part 1194.24 "Video and Multimedia Products" which are of particular importance with regard to all written, graphical or broadcast, video materials or products produced for HHS (to include training). 36 CFR Part 1194.41 outlines the requirements supporting services for products accommodating the communication needs of end-users with disabilities. The deliverables will be provided in Microsoft Word and Adobe PDF formats and compatible with versions currently used at FDA. C.A. Section 508.

This language is applicable to Statements of Work (SOW) or Performance Work Statements (PWS) generated by the Department of Health and Human Services (HHS) that require a contractor or consultant to (1) produce content in any format that could be placed on a Department-owned or Department-funded Web site; or (2) write, create or produce any communications materials intended for public or internal use; to include reports, documents, charts, posters, presentations (such as Microsoft PowerPoint) or video material that could be placed on a Department-owned or Department-funded Web site.

8. SECTION 508 STANDARDS

Electronic and information technology (EIT) supplies and services must comply with Section 508 of the Rehabilitation Act (the Act) of 1973 (29 U.S.C. 794d). See *HHSAR Subpart 339.2*.
Note: Covered solicitations shall include a separate technical evaluation factor specifically addressing Section 508 requirements. Solicitations for supplies and services shall require the submission of a Section 508 Product Assessment Template (see http://www.hhs.gov/web/508 for the template).
Note also: **Prior to making an award decision, the Contracting Officer/Contract Specialist shall forward the Source Selection Evaluation Team's (SSET) assessment of the prospective offeror's responses to the solicitation's Section 508 evaluation factor to the FDA's Section 508 Official or designee for review and approval.**

Section 508 Standards are applicable to this requirement. Electronic and Information Technology (EIT) developed, procured, or maintained for used by federal agencies by a contractor must comply with the standards and regulations. Standards are organized into four subparts:

Subpart A – General (36 CFR §§ 1194.1-1194.5): Describes the types of covered technologies and applications, explains what is exempt, defines terminology, and recognizes that acceptable alternatives that provide equivalent access to and use of a product for people with disabilities are permissible.

Subpart B –Technical Standards (36 CFR § 1194.21-1194.26): Provides criteria specific to various types of technologies and provides performance-based requirements for systems and applications.

- Software applications and operating systems (36 CFR § 1194.21)
- Web-based intranet and internet information and applications (1194.22)
- Telecommunications products (36 CFR § 1194.23)
- Video and multimedia products (36 CFR § 1194.24)
- Self contained, closed products (36 CFR § 1194.25)
- Desktop and portable computers (36 CFR § 1194.26)

Subpart C – Functional Performance Criteria (36 CFR § 1194.31): Provides performance requirements for overall product evaluation and for technologies or components for which there is no specific requirement under the technical standards in Subpart B.

Subpart D –Information, Documentation, and Support (36 CFR § 1194.41): Addresses access to all information, documentation, and support provided to end-users of covered technologies, and identifies formats to be used.

9. SOURCE

Vendors that can provide the required equipment are listed below:

1. Imagine Eyes
   18 rue Charles de Gaulle
   91400 Orsay, France
   Phone: +33 164 86 15 66
   Website: https://www.imagine-eyes.com

2. Canon Inc., Tokyo Japan
   30-2, Shimomaruko 3-chome,
   Tokyo, Japan
   Phone: (81) 3-3758-2111
   https://global.canon

3. EarlySight
   Avenue de Sécheron 15,
   1202 Geneva, SWITZERLAND
   Phone: +41 78 200 71 80

https://www.earlysight.com

4. Physical Sciences Inc., Andover MA
   20 New England Business Center
   Andover, MA, United States
   Phone: (978) 689-0003
   https://www.psicorp.com

5. Robotrak Technologies Co., Ltd.
   6th Floor, Building No.8, 119 Ruanjian Ave, Yuhuatai District, Nanjing
   Jiangsu, China
   Phone: +1 5853199813
   https://www.robotrak.cn/en

10. COMPETITON

☒ Full and Open/Fair Opportunity
☐ Justification for Other than Full and Open Competition/Limited Source Justification
☐ Brand Name or Equal (*specific brand of product or equal to the listed brand*)

Note: Due to the requirement for the contractor/vendor to install all items, all item must be available from one contractor/vendor